Whom to Trust? Trust-Aware Delegation for Decentralized LLM-Agent Networks

Group 21 Progress Report (CSC 8220, Task III, Milestone 2)

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***Abstract*—PPAI (IEEE INFOCOM 2026) lets personalised edge LLM agents delegate queries to more capable peers, but it trusts every peer's self-reported skill and every answer. We show that this assumption is fragile, and we study *trust-aware delegation*: whom to ask, how many peers to ask, and how to combine their answers when some peers lie or collude and every verification costs a network round trip. We built an event-driven simulator grounded in PPAI's measured agent accuracies, together with four baselines. We then designed TADA (Trust-Aware Delegation with Audits), which combines audit-capped Bayesian reputation, anchored agreement and sequential verification. With 30% colluding peers, TADA keeps 0.77 accuracy. Dawid–Skene and majority voting fall to 0.13 or below, and EigenTrust, even with pre-trusted peers, reaches 0.64. TADA also sends 17% fewer probes. A first Q-learning controller for the "ask more or stop" decision roughly halves tail latency under heavy load.**

***Index Terms*—peer-to-peer networks, LLM agents, trust and reputation, collusion, reinforcement learning**

I. Introduction

Edge devices can now host personalised LLM agents, but each device holds only one specialist. PPAI [1] lets a user delegate a query to a peer whose agent is more capable. Queries and agents are matched in a shared prototype space, and a Bayesian load game picks a capable, lightly loaded peer. PPAI, however, *trusts every peer*: capability vectors are self-reported, and answers are accepted at face value. In our simulator, this makes it fragile. With 10% of peers lying about their skill, PPAI's accuracy falls from 0.73 to 0.19; with 10% colluding on a shared wrong answer, it falls to 0.04.

We therefore study trust-aware delegation. Our progress so far: (i) a formulation and threat model; (ii) a discrete-event simulator built on PPAI's agent profiles, with four baselines; (iii) TADA; and (iv) a preliminary reinforcement-learning (RL) controller for the stopping decision.

II. Problem Statement and Formulation

System. There are n peers, each hosting one agent. Every query belongs to one of K skill clusters (PPAI's prototypes). Peer j advertises a capability vector $r_j \in [0,1]^K$ (PPAI's p_j) and answers a cluster-k query correctly with an unknown probability a_{jk}. Each peer is a FIFO server with rate μ_j, and a probe to a remote peer adds a delay d in each direction. A query q has a cluster k_q, an arrival time t_q and a true answer $y_q \in \{1, \dots, L\}$ (multiple choice, $L = 4$).

Decision. The owner's policy π sends probe sets $S_1, S_2, \dots$ in rounds and waits for each round's answers x_S before deciding whether to stop. It then outputs $\hat{y}_q$. It may also send *audits*: gold questions with known answers. The objective is

$$\max_{\pi} \ \mathbb{E}\big[\mathbb{1}\{\hat{y}_q = y_q\}\big] - \gamma\, \mathbb{E}\big[|S(q)| + A(q)\big] - \eta\, \mathbb{E}[T_q], \quad (1)$$

where $|S(q)|$ counts probes, $A(q)$ counts audits, and T_q is the end-to-end latency. There is no trusted third party, no pre-trusted peer, and no ground truth beyond the audit pool.

Threat model. A fraction f of peers advertise $r_{jk} = 0.95$ for every cluster. We consider four behaviours: (T1) *skill liars* answer correctly with probability 0.15; (T2) *colluders* all return the same wrong answer to each query; (T3) colluders *whitewash*, re-joining under a fresh identity once their reputation drops below 0.4 [2]; (T4) colluders *camouflage* by answering correctly with probability 0.3. For now we assume that gold questions are indistinguishable from user queries.

Why AI. Peer reliability is latent and must be inferred from noisy, correlated answers (Bayesian inference, EM). Stopping is a sequential decision under uncertainty, i.e. a small MDP suited to RL. Choosing whom to audit trades exploration against exploitation, as in a bandit problem.

III. Motivation

Why it is interesting. Trust has been studied extensively in P2P file sharing, where a download can be checked against a hash. An LLM answer has no hash: correctness is semantic, honest peers are themselves imperfect (accuracies range from 0.13 to 0.87 in PPAI's Table I), and their errors are correlated. Disagreement is therefore weak evidence of malice, while agreement can be manufactured by a coalition. The problem sits where crowdsourced truth inference, P2P reputation and queueing meet.

Why it is important. Open agent-to-agent networks are emerging, and one wrong medical or financial answer is costly. PPAI itself lists trust as an open problem. Verification also consumes network resources: every extra probe costs a round trip and GPU time on a peer. Deciding how much to verify is therefore a networking resource-allocation problem, not only a security problem.

TABLE I
MILESTONES

When	Milestone	Lead
Oct 13–21	Simulator, baselines, TADA, first RL controller (*done*)	Both
Oct 22–28	Milestone 2: this report and progress slides	Both
Oct 29–Nov 11	Real per-question answers of the 5 models (vLLM [22]); real arrival traces [23], [24]	Trilok
Nov 12–25	Better learned control (per-round credit, audit bandit, adaptive audit rate); stronger attacks	Adele
Nov 26–final	Final experiments, final report ($\geq$6 pages), 25-minute talk	Both

IV. Related Work

LLM routing and ensembles. RouterDC [3], RouteLLM [4] and KABB [5] route each query to a model on a single trusted platform. Self-consistency [6], "More Agents" [7] and LLM-Blender [8] show that aggregating answers improves accuracy, but they assume every member is honest. PPAI [1] decentralises routing but keeps the honesty assumption.

P2P reputation. EigenTrust [9] computes global trust by power iteration anchored on pre-trusted peers. PeerTrust [10] and the Beta reputation system [11] aggregate feedback about transactions. Surveys [12], [13] catalogue attacks such as whitewashing [2] and Sybil identities [14]. All of these assume the outcome of a transaction can be observed; we have no such feedback.

Truth inference in crowdsourcing. Dawid–Skene [15], GLAD [16] and Karger et al. [17] infer worker reliability without labels, but they assume workers err independently, which collusion violates. Gold questions [18] inspire our audits.

Byzantine fault tolerance and verifiable inference. BFT [19], [20] needs deterministic replicas, whereas honest LLMs legitimately disagree. Zero-knowledge proofs of inference [21] are orders of magnitude too slow for interactive edge queries.

Gap. We are not aware of work that combines latent-reliability inference with adaptive probe allocation under collusion in a delay-sensitive network.

V. Plan of Action

Table I lists our milestones. **Division of work.** T. Padhi proposed the core direction of modelling disagreement between peers to assign trust. He leads threat models, the literature review, the data pipelines and the evaluation. A. Chinda leads the simulator, TADA and the RL controller. Both of us write and present.

VI. Research Progress

A. Simulator, data and baselines

We built a discrete-event simulator with Poisson arrivals, FIFO peer queues and network delay. Policies are written as processes that send probes and wait for the replies. It has 50 peers, each an instance of one of PPAI's five agents (Qwen2.5-7B, DeepSeek-R1-Distill-7B, Nemotron-Code-7B, HuatuoGPT-o1-7B, Llama-3-8B). Their accuracies on MMLU, GSM8K, MedQA, ARC-C and AGIEval are taken from PPAI's Table I, plus per-peer noise. When honest peers are wrong, they pick a shared distractor with probability 0.5, so their errors are correlated. Other settings: service rates of 15–25 queries/s, a one-way delay of 5 ms, an arrival rate $\lambda = 50$/s, 5000 queries per run, and 5 seeds. **Baselines**: PPAI's argmax; majority vote over the 5 most "relevant" peers; Dawid–Skene EM (one-coin model, sliding window); and EigenTrust, which we give 3 pre-trusted honest peers, an advantage TADA does not get.

B. TADA

Per query, TADA (1) shortlists peers by reported relevance ($r_{jk} \geq 0.3$, PPAI's scoring stage); (2) ranks them by *estimated* accuracy $\hat{a}_{jk}$ minus β times their queued work, so reported skill is only a filter and lying about it buys nothing; (3) asks the top 3 peers and, if the posterior of the best answer is below $\tau = 0.9$, asks 2 more; and (4) with probability 0.15, sends a gold question from PPAI's own per-prototype question set Q_k, either to a peer it just used or to the shortlisted peer whose audit record is most uncertain. Each peer keeps Beta counts per cluster [11] for agreement, $(\alpha_{jk}, \beta_{jk})$, and for audits, (g^+_{jk}, g^-_{jk}). Agreement may never claim more than the audit record supports:

$$\hat{a}_{jk} = \min\Big\{ \tfrac{\alpha_{jk}}{\alpha_{jk}+\beta_{jk}},\ U(g^+_{jk}, g^-_{jk}),\ U(g^+_{j\cdot}, g^-_{j\cdot}) + \delta \Big\}, \quad (2)$$

where $U = \hat{p} + z\sqrt{\hat{p}(1-\hat{p})/(n+1)}$ is an upper confidence bound on the audit success rate $\hat{p}$ over n audits, $z = 1.28$ and $\delta = 0.15$. The last term pools a peer's audits over all clusters. Answers are combined with the one-coin posterior

$$P(y = \ell \mid x_S) \ \propto\ \prod_{j \in S} \hat{a}_{jk}^{\mathbb{1}[x_j = \ell]} \Big(\frac{1 - \hat{a}_{jk}}{L - 1} \Big)^{\mathbb{1}[x_j \neq \ell]}. \quad (3)$$

Anchored agreement: agreement counts are updated only if the same answer still wins, with posterior ≥ 0.5, when each peer is weighted by its *audit-only* lower bound, floored at chance $(1/L)$. Unaudited colluders then carry no weight and cannot bootstrap their own trust.

C. Initial findings

(1) Trust-blind and independence-based methods fail (Fig. 1a,b). PPAI is captured by any liar. Dawid–Skene handles independent liars (0.77 at 30%) but collapses under collusion (0.01), because colluders look like a reliable majority. **(2) TADA is robust without pre-trusted peers.** It reaches 0.77 with 30% colluders and 0.72 with 50%, against 0.64 and 0.62 for EigenTrust. Only 4% of the answers it uses come from adversaries (13% for EigenTrust). It survives whitewashing (0.75) and camouflage (0.76), and it learns quickly: accuracy rises from 0.53 to 0.83 within the first 30% of a run. **(3) Audits and anchoring do the work** (Table II). Without audits,

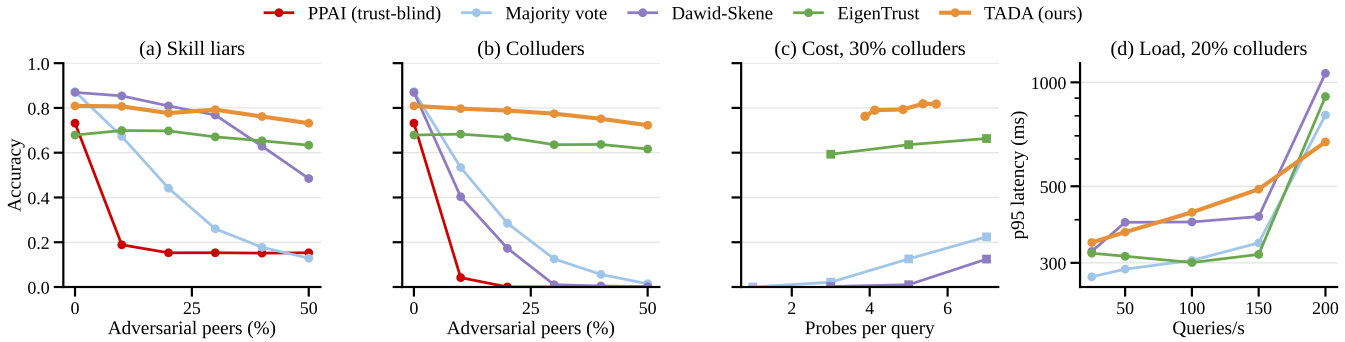


Fig. 1. (a,b) Accuracy as the share of adversarial peers grows. (c) Accuracy vs. probes per query with 30% colluders (committee size or τ swept). (d) p95 latency vs. arrival rate with 20% colluders. Means over 5 seeds; EigenTrust is given 3 pre-trusted peers, TADA none.

TABLE II
ACCURACY WITH 30% ADVERSARIES OF EACH KIND (WW: WHITEWASHING, CAM.: CAMOUFLAGE); PROBES PER QUERY AND P95 LATENCY (MS) WITH COLLUDERS. ROWS MARKED − REMOVE ONE TADA COMPONENT.

Method	None	Liars	Coll.	+WW	+Cam.	Probes	p95
TADA (full)	0.81	**0.79**	**0.77**	**0.75**	**0.76**	4.2	376
− anchoring	0.82	0.78	0.63	0.47	0.67	4.1	370
− audits	0.76	0.57	0.42	0.42	0.61	4.8	432
− pooled audit cap	0.81	0.79	0.76	0.74	0.76	4.2	373
− sequential	0.82	0.79	0.77	0.75	0.76	5.1	276
EigenTrust (m=5)	0.68	0.67	0.64	0.67	0.66	5.0	301
Dawid–Skene (m=5)	**0.87**	0.77	0.01	0.01	0.34	5.0	333
Majority vote (m=5)	**0.87**	0.26	0.13	0.13	0.31	5.0	268

accuracy under collusion drops to 0.42. Without anchoring, it drops to 0.63, and to 0.47 under whitewashing. The pooled audit cap adds little once anchoring is in place. **(4) Verification is a network cost.** Sequential verification saves 18% of probes at equal accuracy but raises p95 latency from 276 to 376 ms. Under heavy load the saving pays off: at 200 queries/s, TADA has the lowest p95 latency (672 ms vs. 804–1062 ms; Fig. 1d). **(5) The price of scepticism**: with no attackers, TADA is 6 points below a full 5-peer vote.

D. Preliminary RL controller

We replaced the fixed threshold τ with tabular Q-learning [25]. The state is (confidence bin, round, load bin, cluster), and the actions are {stop, ask 2 more}. The reward is correctness minus costs per probe and per second of latency. The controller is trained in simulation, where ground truth is known, on seeds disjoint from the evaluation seeds. With 30% colluders at 150 queries/s, it learns to stop earlier: p95 latency falls from 576 to 310 ms and probes from 4.3 to 3.3, for -1.4 accuracy points. With the latency penalty removed, it matches the best threshold (0.79) but does not beat it. Binary, high-variance rewards limit it so far.

E. Challenges and how we addressed them

(a) In our first answer model, wrong answers never agreed with each other, which made majority voting look almost perfect. We switched to multiple-choice answers with a shared distractor. **(b)** Agreement-only trust was captured by colluders (0.003 accuracy at 30%). Capping trust by audits fixed the steady state, and anchoring fixed the warm-up (+14 points). **(c)** Our first queue model reserved servers for future verification rounds, producing spurious 12 s latencies. We rebuilt the simulator as a discrete-event simulation. **(d)** Asking a single peer first caused up to 3 sequential rounds. Asking 3 first caps each query at two rounds and cuts p95 latency by about 25%.

F. Algorithms to develop or refine

(i) Per-round credit assignment and audit-calibrated rewards for the RL controller. (ii) A contextual bandit for choosing whom to audit, and an adaptive audit rate. (iii) Collusion detection from answer correlations. (iv) Stronger adversaries that try to detect gold questions, and Sybil swarms [14]. (v) Replacing the synthetic answer model with real per-question answers from the five models. Code, tests, and the script that regenerates every number in this report are in the supplementary material.

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